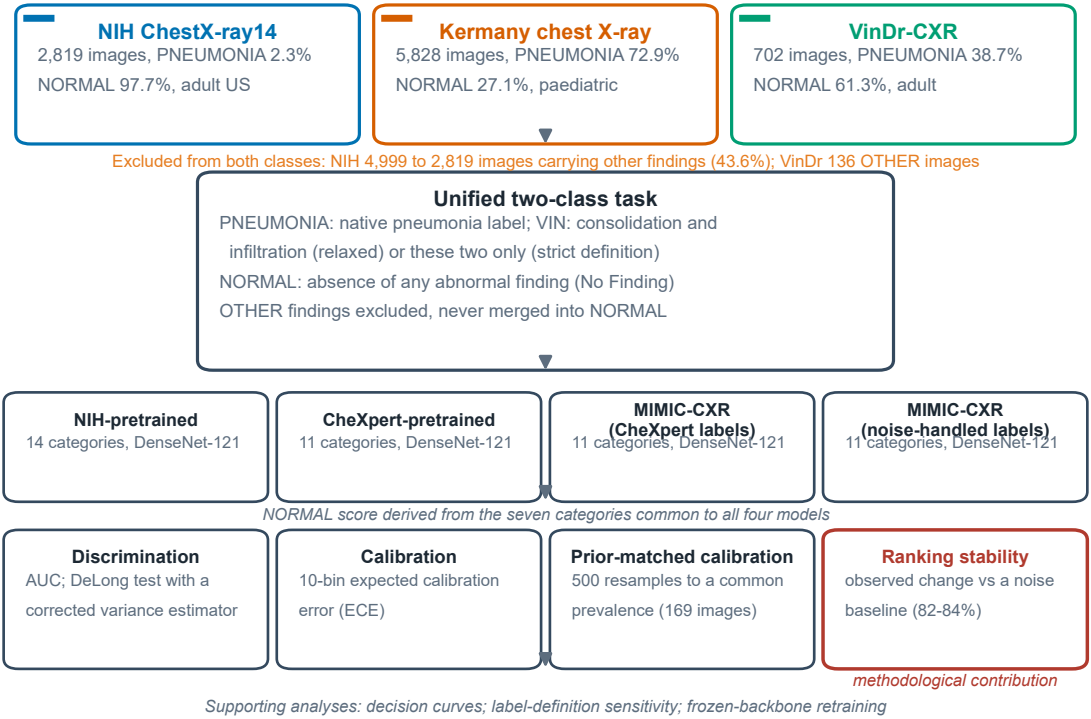

NIH ChestX-ray14
2,819 images, PNEUMONIA 2.3%
NORMAL 97.7%, adult US
Kermany chest X-ray
5,828 images, PNEUMONIA 72.9%
NORMAL 27.1%, paediatric
VinDr-CXR
702 images, PNEUMONIA 38.7%
NORMAL 61.3%, adult
Excluded from both classes: NIH 4,999 to 2,819 images carrying other findings (43.6%); VinDr 136 OTHER images
Unified two-class task
PNEUMONIA: native pneumonia label; VIN: consolidation and infiltration (relaxed) or these two only (strict definition)
NORMAL: absence of any abnormal finding (No Finding)
OTHER findings excluded, never merged into NORMAL
NIH-pretrained
14 categories, DenseNet-121
CheXpert-pretrained
11 categories, DenseNet-121
MIMIC-CXR (CheXpert labels)
11 categories, DenseNet-121
MIMIC-CXR (noise-handled labels)
11 categories, DenseNet-121
NORMAL score derived from the seven categories common to all four models
Discrimination
AUC; DeLong test with a corrected variance estimator
Calibration
10-bin expected calibration error (ECE)
Prior-matched calibration
500 resamples to a common prevalence (169 images)
Ranking stability
observed change vs a noise baseline (82-84%)
methodological contribution
Supporting analyses: decision curves; label-definition sensitivity; frozen-backbone retraining